STATISTICS AND DATA SCIENCE 188

TOPIC DIVIDER 07

Sampling Distributions

Sample means, sample proportions, the Central Limit Theorem, differences between independent sample means, and finite-population correction.

I. TOPIC MAP

Identify the random object → **Choose X, $\bar{X}$, p, or $\bar{X}_1$-$\bar{X}_2$** → **Find the centre** → **Calculate the standard error** → **Check distribution conditions** → **Standardise** → **Find the probability or boundary** → **Interpret in context**

II. KEY CONCEPTS

- **Population distribution**
- **Sample**
- **Statistic**
- **Sampling distribution**
- **Sample mean**
- **Sample proportion**
- **Unbiased estimator**
- **Standard error**
- **Central Limit Theorem**
- **Difference between independent sample means**
- **Finite-population correction**

III. MAIN DISTINCTIONS / RULES

1 - X versus $\bar{X}$

X is one observation; $\bar{X}$ is a random-sample average.

Use σ for X and $\sigma/\sqrt{n}$ for $\bar{X}$.

2 - SD versus SE

σ describes variability among individuals; SE describes variability among possible statistics.

Larger n reduces SE, not the centre.

3 - Normal versus CLT

A valid independent random sample from a normal population gives normal $\bar{X}$ for any n.

For non-normal populations, sufficiently large n may justify the CLT.

4 - Ordinary SE versus FPC

Use ordinary SE with replacement or a negligible sampling fraction.

Without replacement from finite N, apply FPC when $n/N > 0.05$.

IV. EXAM FOCUS

- Identify whether the question concerns X, $\bar{X}$, p or $\bar{X}_1$-$\bar{X}_2$.
- Write the centre and standard error before standardising.
- Check normality, CLT, proportion or independence conditions.
- Translate below, above, between and inverse wording into the correct normal region.
- Interpret the final probability or boundary in the context of the sample statistic.

V. COMMON TRAPS

- Using σ instead of $\sigma/\sqrt{n}$ for a sample mean.
- Dividing by n instead of $\sqrt{n}$.
- Confusing variance with standard deviation.
- Saying the population becomes normal under the CLT.
- Using the wrong tail or forgetting the complement.
- Forgetting FPC above 5% without replacement.

BEFORE MOVING ON, I SHOULD KNOW...

- ☐ I can distinguish X from $\bar{X}$.
- ☐ I can decide whether the normal model or CLT applies.
- ☐ I can solve sample-proportion problems and check both conditions.
- ☐ I can decide whether finite-population correction is required.
- ☐ I can calculate and explain a standard error.
- ☐ I can solve below, above, between and inverse mean problems.
- ☐ I can calculate probabilities for two independent sample means.
- ☐ I can explain unbiasedness and the effect of increasing n.

SDS188 · CHAPTER 7

Sampling Distributions Formula Sheet

Centre · standard error · conditions · probability

Population parameters: μ, σ, π | Sample statistics: $\bar{X}$, p

Question concerns...	Random object	Centre	Spread
One observation	X	μ	σ
Average of n observations	$\bar{X}$	μ	$\frac{\sigma}{\sqrt{n}}$
Sample proportion	p	π	$\sqrt{\frac{\pi(1-\pi)}{n}}$
Difference between two independent sample means	$\bar{X}_1-\bar{X}_2$	$\mu_1-\mu_2$	$\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}$

Identify object → write centre → write SE → check conditions → draw region → standardise → calculate area → interpret

1. POPULATION FORMULAS

$$\mu=\frac{\sum_{i=1}^{N}X_i}{N} \qquad \sigma^2=\frac{\sum_{i=1}^{N}(X_i-\mu)^2}{N} \qquad \sigma=\sqrt{\frac{\sum_{i=1}^{N}(X_i-\mu)^2}{N}}$$

Use N for a complete listed population, not n-1.

2. GENERAL EXACT SAMPLING DISTRIBUTIONS

$$\mu_T=\sum tP(t) \qquad \sigma_T^2=\sum(t-\mu_T)^2P(t)$$

$$\sigma_T=\sqrt{\sum(t-\mu_T)^2P(t)}$$

$$P(T=t)=\frac{\text{frequency of } t}{\text{total number of equally likely samples}}$$

T is the statistic; t is one possible value. Use exact formulas when all values and probabilities are listed.

$$\binom{N}{n}=\frac{N!}{n!(N-n)!}$$

ordered, with replacement: N^n

example: $4^2=16$

Tutorial 7 Q4: combinations count unordered samples without replacement.

3. INDIVIDUAL OBSERVATION AND SAMPLE MEAN

ONE INDIVIDUAL OBSERVATION (NORMAL X)

$$Z=\frac{X-\mu}{\sigma}$$

One person, item, time or observation: do not divide σ by √n.

SAMPLING DISTRIBUTION OF THE MEAN

$$E(\bar{X})=\mu_{\bar{X}}=\mu \qquad \sigma_{\bar{X}}^2=\frac{\sigma^2}{n} \qquad \sigma_{\bar{X}}=\frac{\sigma}{\sqrt{n}}$$

$$X\sim N(\mu,\sigma^2) \implies \bar{X}\sim N(\mu,\sigma^2/n)$$

Notation on this sheet: N(mean, variance). Random independent sample required.

$$Z=\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \qquad \bar{X}=\mu+Z\frac{\sigma}{\sqrt{n}}$$

$$\bar{X}_L=\mu+z_L\frac{\sigma}{\sqrt{n}} \;;\; \bar{X}_U=\mu+z_U\frac{\sigma}{\sqrt{n}} \qquad \mu\pm z^*\frac{\sigma}{\sqrt{n}}$$

Ordinary mean SE - use unless the finite-population correction applies.

4. CENTRAL LIMIT THEOREM

$$\bar{X}\approx N\left(\mu,\frac{\sigma^2}{n}\right)$$

Use when the CLT provides an adequate normal approximation.

Course guide: about n≈15 if fairly symmetric; about n≈30 for most populations; use greater caution with strong skewness or outliers.

5. SAMPLE PROPORTION

$$p=\frac{X}{n} \qquad \mu_p=\pi \qquad \sigma_p=\sqrt{\frac{\pi(1-\pi)}{n}}$$

Ordinary proportion SE - use unless FPC applies. X is the success count.

$$p\approx N\left(\pi,\frac{\pi(1-\pi)}{n}\right) \;;\; n\pi\geq 5,\quad n(1-\pi)\geq 5$$

The second N parameter is the variance. Known π: use π in theoretical SE.

$$Z=\frac{p-\pi}{\sqrt{\pi(1-\pi)/n}} \qquad \hat{\sigma_p}=\sqrt{\frac{p(1-p)}{n}}$$

Unknown π and estimated SE requested: use observed p. No continuity correction.

6. DIFFERENCE BETWEEN TWO INDEPENDENT SAMPLE MEANS

$$\mu_{\bar{X}_1-\bar{X}_2}=\mu_1-\mu_2$$

$$\sigma_{\bar{X}_1-\bar{X}_2}^2=\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}$$

$$\sigma_{\bar{X}_1-\bar{X}_2}=\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}$$

$$\bar{X}_1-\bar{X}_2\sim N\left(\mu_1-\mu_2,\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}\right)$$

$$Z=\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{\sqrt{\sigma_1^2/n_1+\sigma_2^2/n_2}}$$

Applies with independence within/between random samples, known population SDs, and normal populations or sufficiently large samples.

Expected difference is $\mu_1-\mu_2$, not zero by default.

7. FINITE-POPULATION CORRECTION

$$f=\frac{n}{N} \qquad FPC=\sqrt{\frac{N-n}{N-1}}$$

Use only without replacement from finite N when n/N > 0.05.

$$\sigma_{\bar{X}}=\frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$$

$$\sigma_p=\sqrt{\frac{\pi(1-\pi)}{n}}\sqrt{\frac{N-n}{N-1}}$$

FPC changes spread only; it does not change μ or π.

8. NORMAL PROBABILITY AND INVERSE RULES

$$\Phi(z)=P(Z\leq z) \mid P(Z<z)=\Phi(z) \mid P(Z>z)=1-\Phi(z)$$

$$P(a<Z<b)=\Phi(b)-\Phi(a)$$

$$P(Z>z)=P(Z<-z), \quad \Phi(-z)=1-\Phi(z)$$

$z=\Phi^{-1}(A)$ A = cumulative area to the left

boundary = centre + z·SE

$$p_L=\pi+z_L\sigma_p \qquad p_U=\pi+z_U\sigma_p$$

$$\alpha=1-C \qquad \text{tail each}=\frac{\alpha}{2} \qquad z^*=\Phi^{-1}\left(\frac{1+C}{2}\right)$$

Central area	Tail each side	z^*
90%	0.05	1.6449
95%	0.025	1.9600
99%	0.005	2.5758

Sampling-distribution boundary references only; not confidence-interval teaching.

9. TUTORIAL 7 BRIDGE - EXPONENTIAL WAITING TIME

$$\mu_X=\frac{1}{\lambda} \qquad \sigma_X=\frac{1}{\lambda} \qquad P(X>t)=e^{-\lambda t}$$

$$\mu_{\bar{X}}=\frac{1}{\lambda} \qquad \sigma_{\bar{X}}=\frac{1/\lambda}{\sqrt{n}}$$

Rate and time must use consistent units.

10. FINAL CHECKS

- ✓ Object and centre
- ✓ Conditions and tail
- ✓ Correct SE / FPC
- ✓ Full precision and context

TRAP STRIP Variance 36 → σ=6 • Percent → decimal • Mean SE divides by √n • Right tail = 1-Φ(z); between = upper-lower CDF • Keep order and full precision • No FPC with replacement.